



AREA CARTOGRAM

ALSO KNOWN AS Contiguous cartogram, density-equalising map

C H R T S

DISTORTIONS

REPRESENTATION DESCRIPTION

An area cartogram displays the quantitative values associated with distinct, definable spatial regions on a map. Each geographic region is represented by a polygonal area based on its outline shape with the collective regional shapes forming the entire landscape. Quantitative values are represented by proportionately distorting (inflating or deflating) the relative size of and, to some degree, shape of the respective regional areas. Traditionally, area cartograms strictly aim to preserve the neighbourhood relationships between different regions. Attributes of colour are often used to represent the quantitative measurements and/or to associate the region with a categorical classification. Area cartograms require the reader to be relatively familiar with the original size and shape of regions in order to be able to establish the degree of relative change in their proportions.

EXAMPLE Mapping the measures of climate change responsibility compared with vulnerability across all countries.

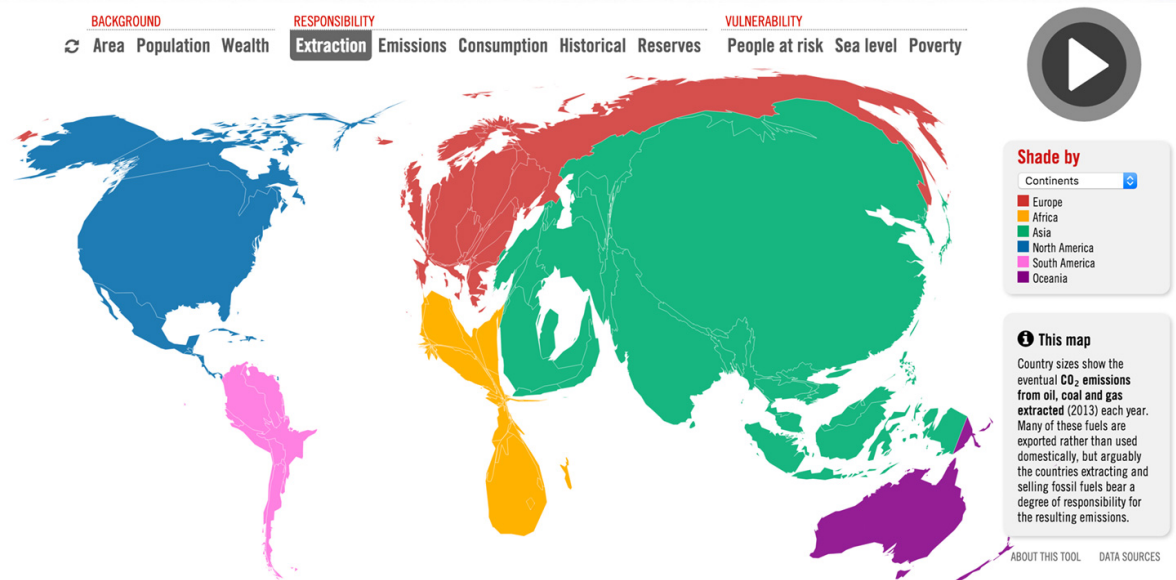


Figure 6.50 The Carbon Map, by Duncan Clark and Robin Houston (Kiln)

PRESENTATION TIPS

INTERACTIVITY: Animated sequences enabled through interactive controls can help to better identify instances and degrees of change, but usually only over a small set of regions and only if the change is relatively smooth and sustained. Manual animation will help to provide more control over the experience. Selectable tooltips to view quantitative values and category or location labels for any region on the display may also prove useful.

ANNOTATION: Directly labelling the regional areas with geographic details and the value they hold is likely to lead to too much clutter. As it is difficult to assess the degree of distortion and, indeed, often to identify the regions themselves, it can be useful to present a thumbnail view of the undistorted original geographic layout to help readers orient themselves with the changes. Additionally, a limited number of regional labels might be included to provide direct spatial context and orientation. Any colours used must be explained through the inclusion of a legend.

COLOUR: The outline colour and stroke width for each spatial area should be distinguishable enough to define the shape but not so prominent as to dominate. Usually, a light-grey or white-coloured stroke will suffice.

VARIATIONS & ALTERNATIVES

Unlike contiguous cartograms, non-contiguous cartograms tend to preserve the shape of the individual polygons but modify the size and the neighbouring connectivity to other adjacent regional polygon areas. The best alternative ways of showing similar data would be to consider using the 'choropleth map' or 'Dorling cartogram'.